

Daniel Jilek

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Education

Northeastern University

Master of Science, Computer Science

Expected Graduation: December 2026

GPA: 3.55

University of Texas at Austin

Bachelor of Science, Neuroscience

May 2022

GPA: 3.72

Skills

- Languages: Python, C, C++, Java, JavaScript (ES6+), SQL
- Frameworks/Libraries: React, Node.js, Express.js, PyTorch, OpenCV
- Tools/DevOps: Git, GitHub, Docker, MongoDB, Postman
- Core Concepts: Object-Oriented Programming (OOP), RESTful APIs, Real-time Networking (WebSockets), Deep Learning, Image Processing

Relevant Coursework

Object-Oriented Programming, Computer Vision, Algorithms, Data Structures.

Projects

Libro de los Juegos | *React, Node.js, ExpressJS, MongoDB, Socket.io*

- Engineered a full-stack web application hosting real-time multiplayer tabletop games.
- Implemented WebSockets (Socket.io) to facilitate low-latency, bidirectional communication between clients for synchronized gameplay.
- Designed a scalable REST API and managed user state and game history using MongoDB.

YOLO V1 Pre-training Classifier | *Python, PyTorch, Torchvision*

- Architected a convolutional neural network based on the YOLO v1 pre-training classifier architecture for image classification tasks.
- Optimized model performance through custom data augmentation and hyperparameter tuning using the iNaturalist 2017 mammal dataset.
- Leveraged PyTorch for GPU-accelerated training and evaluation, achieving 41% top-5 accuracy on the validation set.

Realtime 2-D Object Recognition | *C++, OpenCV*

- Developed a real-time computer vision tool to identify and classify objects within live video streams.
- Implemented classical feature extraction techniques to improve recognition stability under varying light conditions.
- Compared performance of classical feature based object recognition with ResNet feature vectors

Work Experience

Research Associate I

MassBiologics

Boston, Massachusetts

August 2022 – May 2023

- Produced, purified, and analyzed antibody and nanobody candidates for therapeutic treatment against different variants of COVID-19.
- Analyzed and interpreted data using GraphPad Prism and presented conclusions to the department.